Raphael Kaíque Dias Santos

Backend Software Engineer

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PROFESSIONAL SUMMARY

Backend-focused Software Engineer with professional software development experience since 2020, specializing in PHP, Symfony, TypeScript/Node.js, NestJS, Go, Gin, PostgreSQL, MongoDB, GraphQL, RESTful APIs, Cloud (AWS/GCP) and software architecture. Experienced across the software development lifecycle, including requirements analysis, system design, implementation, integration, testing, deployment, troubleshooting, and production support. Background spans IoT, telecommunications, AI/computer vision, remote teleoperation, autonomous vehicles, real-time communication, distributed systems, and Linux infrastructure.

Comfortable working across backend services, APIs, databases, cloud services, networking, and infrastructure, with experience integrating software with operational and industrial systems. Strong foundation in system integration, scalable architecture, and maintainable software engineering practices.

CORE TECHNOLOGIES

Backend: PHP, Symfony, TypeScript, Node.js, NestJS, Go, Gin, Java, Spring Boot

Databases: PostgreSQL, MySQL, Microsoft SQL Server, SQLite, MongoDB, Redis, Neo4j, Oracle

APIs & Architecture: REST/RESTful APIs, SOAP, GraphQL, gRPC, WebSockets, WebRTC, Microservices, Event-Driven Architecture, DDD, SOLID, Design Patterns, Hexagonal Architecture, Clean Architecture, MVC, System Design, TDD

Infrastructure & Cloud: Linux, Shell Scripting, NGINX, Apache, Docker/Compose, Git, GitHub, GitLab, Jenkins CI/CD, SSH, Virtualization, AWS, GCP, Azure

Messaging & Real-Time: RabbitMQ, MQTT/Mosquitto, TCP/UDP, RTP/RTSP, SIP, ONVIF, cURL, JSON, JWT

Additional: JavaScript, Vue, Angular, C#/ASP.NET/.NET, C/C++, Python, Qt/QML, IoT, Telemetry, CAN/CANopen, OpenCV, FFmpeg/GStreamer

PROFESSIONAL EXPERIENCE

SEESTEC Engenharia e Tecnologia — Development Analyst | Jan 2026 – Present

- Design, develop, test, deploy, and maintain software solutions across the software development lifecycle, from requirements analysis and system architecture to production support.
- Develop and evolve software solutions for AI, IoT, remote teleoperation, and autonomous vehicle applications, working across backend systems, real-time communication, telemetry, computer vision, and distributed infrastructure.
- Design and implement APIs, web services, backend routines, and integrations with internal and third-party systems using PHP/Symfony, TypeScript/Node.js/NestJS, Linux, NGINX, Docker/Compose, Git, Shell Scripting, MongoDB, Redis, MySQL, and SQLite.
- Develop and integrate real-time communication systems involving WebSockets, WebRTC, RTP/RTSP, SIP, TCP/UDP, MQTT, and ONVIF.
- Work with AI and computer vision environments using NVIDIA Jetson, YOLO, OpenCV, and Tesseract, including integration of LLM, RAG, and AI-agent capabilities.
- Perform unit, integration, and system testing, troubleshooting, root-cause analysis, performance optimization, technical documentation, and production support.

ARQIA Telecom & IoT Latam — Development Analyst | Sep 2025 – Jan 2026

- Developed and maintained software solutions for IoT, MVNO, and VoIP environments.
- Developed REST/SOAP APIs, web services, and system integrations using PHP, JavaScript, MySQL, C#/.NET, ASP.NET, and Microsoft SQL Server.
- Worked with Microsoft SQL Server, T-SQL, Entity Framework/LINQ, and Azure services including Logic Apps, Functions, Storage, and API Management.
- Performed unit, integration, and system testing, bug fixing, security practices, and performance optimization.
- Collaborated with analysts, testers, designers, and developers in Scrum and Kanban environments.

SITECH Brazil — Technical Analyst | Nov 2023 – Oct 2024

- Developed software solutions for teleoperation imaging and remote control of autonomous vehicles using C/C++, JavaScript, React, Node.js, Python, MySQL, Qt/QML, and MQTT/Mosquitto.
- Developed, tested, deployed, and maintained software environments and database systems.
- Administered database servers, supported Linux/server infrastructure, and performed network diagnostics.

- Worked with telemetry networks and CAN/CANopen in industrial systems, integrating software, networking, and operational technologies.

Sotreq | CAT Latam — Commercial Assistant / Automation & Data | Sep 2020 – Nov 2023

- Developed process-automation and data-analysis tools using Visual Basic, ABAP, and SQLite.
- Built KPIs and interactive dashboards using Power BI, DAX, Excel, and VBA to support commercial analysis and decision-making.

EDUCATION

Postgraduate Degree in Software Architecture | FIAP — Faculdade de Informática e Administração Paulista | Dec 2026 – Jan 2028

Coursework focused on software engineering, application development, information technology, artificial intelligence, databases, cloud computing, and software systems.

Technology Degree in DevOps | UNIDERP — Universidade Anhanguera | Feb 2021 – Jul 2023

Focus on DevOps engineering, software development, infrastructure automation, CI/CD, cloud and networking, monitoring and observability, containerization and orchestration, security, and software delivery practices.

ENGINEERING PRACTICES

SOLID • DDD • TDD • Design Patterns • Clean Architecture • Hexagonal Architecture • MVC • Microservices • Event-Driven Architecture • System Design • Design Documentation • Agile/Scrum/Kanban • CI/CD • Docker • Git

LANGUAGES

Portuguese — Native | English — Professional Technical Proficiency